

Exercise 2C: Simplification of Fractions

Step-by-Step Solutions using the BODMAS Rule

Comprehensive Guide to Fractional Arithmetic and Operations

The BODMAS Rule for Fractions

When simplifying numerical expressions containing fractions, operations must be performed strictly in the following order:

1. **B (Brackets):** Perform operations enclosed in brackets first. The order of brackets is: parentheses $()$, braces $\{\}$, and brackets $[\]$.
2. **O (Of):** The word “of” acts as a high-priority multiplication (e.g., $\frac{a}{b}$ of $\frac{c}{d} = \frac{a \times c}{b \times d}$), evaluated before division.
3. **D (Division):** Replace division with multiplication by the reciprocal of the divisor ($\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$).
4. **M (Multiplication):** Multiply numerators and denominators respectively.
5. **A (Addition):** Convert fractions to a common denominator (using the LCD) and add.
6. **S (Subtraction):** Convert fractions to a common denominator and subtract.

Question 1: Simplify

Expression: $1\frac{2}{3} \div \frac{5}{6}$ of $\frac{24}{25}$

Important Note on Printing Variations

Depending on your textbook printing version, the operator between $1\frac{2}{3}$ and $\frac{5}{6}$ may be printed as division ($\div$) or addition ($+$). We provide rigorous step-by-step solutions for both cases to guarantee you get the exact correct solution for your class.

Case A: If the expression is $1\frac{2}{3} \div \frac{5}{6}$ of $\frac{24}{25}$ (Standard BODMAS)

1. Convert Mixed Numbers to Improper Fractions:

$$1\frac{2}{3} = \frac{(1 \times 3) + 2}{3} = \frac{5}{3}$$

The expression becomes:

$$\frac{5}{3} \div \frac{5}{6} \text{ of } \frac{24}{25}$$

2. Perform the “Of” Operation First: Evaluate $\frac{5}{6}$ of $\frac{24}{25}$:

$$\frac{5}{6} \text{ of } \frac{24}{25} = \frac{5}{6} \times \frac{24}{25} = \frac{\overset{1}{\cancel{5}} \times \overset{4}{\cancel{24}}}{\underset{1}{\cancel{6}} \times \underset{5}{\cancel{25}}} = \frac{1 \times 4}{1 \times 5} = \frac{4}{5}$$

The expression simplifies to:

$$\frac{5}{3} \div \frac{4}{5}$$

3. Perform Division (Multiply by Reciprocal): The reciprocal of $\frac{4}{5}$ is $\frac{5}{4}$.

$$\frac{5}{3} \div \frac{4}{5} = \frac{5}{3} \times \frac{5}{4} = \frac{5 \times 5}{3 \times 4} = \frac{25}{12} = 2\frac{1}{12}$$

Final Answer (Case A): $2\frac{1}{12}$

Case B: If the expression is $1\frac{2}{3} + \frac{5}{6}$ of $\frac{24}{25}$

1. Perform the “Of” Operation First: As calculated above, $\frac{5}{6}$ of $\frac{24}{25} = \frac{4}{5}$.

2. Perform Addition: We now evaluate:

$$1\frac{2}{3} + \frac{4}{5} = \frac{5}{3} + \frac{4}{5}$$

The Least Common Denominator (LCD) of 3 and 5 is 15:

$$\frac{5}{3} = \frac{5 \times 5}{3 \times 5} = \frac{25}{15}, \quad \frac{4}{5} = \frac{4 \times 3}{5 \times 3} = \frac{12}{15}$$

Adding the numerators:

$$\frac{25}{15} + \frac{12}{15} = \frac{25 + 12}{15} = \frac{37}{15} = 2\frac{7}{15}$$

Final Answer (Case B): $2\frac{7}{15}$

Question 2: Simplify

Expression: $\frac{1}{3}$ of $4\frac{2}{3} \div 2\frac{1}{3} \times 1\frac{1}{2}$

1. Convert Mixed Numbers to Improper Fractions:

$$4\frac{2}{3} = \frac{(4 \times 3) + 2}{3} = \frac{14}{3}, \quad 2\frac{1}{3} = \frac{(2 \times 3) + 1}{3} = \frac{7}{3}, \quad 1\frac{1}{2} = \frac{(1 \times 2) + 1}{2} = \frac{3}{2}$$

The expression becomes:

$$\frac{1}{3} \text{ of } \frac{14}{3} \div \frac{7}{3} \times \frac{3}{2}$$

2. Perform the “Of” Operation First: Evaluate $\frac{1}{3}$ of $\frac{14}{3}$:

$$\frac{1}{3} \text{ of } \frac{14}{3} = \frac{1}{3} \times \frac{14}{3} = \frac{1 \times 14}{3 \times 3} = \frac{14}{9}$$

The expression becomes:

$$\frac{14}{9} \div \frac{7}{3} \times \frac{3}{2}$$

3. Perform Division: The reciprocal of $\frac{7}{3}$ is $\frac{3}{7}$.

$$\frac{14}{9} \div \frac{7}{3} = \frac{14}{9} \times \frac{3}{7} = \frac{\overset{2}{\cancel{14}} \times \overset{1}{\cancel{3}}}{\underset{3}{\cancel{9}} \times \underset{1}{\cancel{7}}} = \frac{2 \times 1}{3 \times 1} = \frac{2}{3}$$

The expression becomes:

$$\frac{2}{3} \times \frac{3}{2}$$

4. Perform Multiplication:

$$\frac{2}{3} \times \frac{3}{2} = \frac{\overset{1}{\cancel{2}} \times \overset{1}{\cancel{3}}}{\underset{1}{\cancel{3}} \times \underset{1}{\cancel{2}}} = 1$$

Final Answer: 1

Question 3: Simplify

Expression: $2\frac{1}{4} + 1\frac{1}{6} - 1\frac{2}{3} \div 2\frac{2}{3}$ of $3\frac{3}{4}$

1. Convert Mixed Numbers to Improper Fractions:

$$2\frac{1}{4} = \frac{9}{4}, \quad 1\frac{1}{6} = \frac{7}{6}, \quad 1\frac{2}{3} = \frac{5}{3}, \quad 2\frac{2}{3} = \frac{8}{3}, \quad 3\frac{3}{4} = \frac{15}{4}$$

The expression becomes:

$$\frac{9}{4} + \frac{7}{6} - \frac{5}{3} \div \frac{8}{3} \text{ of } \frac{15}{4}$$

2. Perform the “Of” Operation First: Evaluate $\frac{8}{3}$ of $\frac{15}{4}$:

$$\frac{8}{3} \text{ of } \frac{15}{4} = \frac{8}{3} \times \frac{15}{4} = \frac{\overset{2}{\cancel{8}} \times \overset{5}{\cancel{15}}}{\underset{1}{\cancel{3}} \times \underset{1}{\cancel{4}}} = \frac{2 \times 5}{1 \times 1} = 10$$

The expression becomes:

$$\frac{9}{4} + \frac{7}{6} - \frac{5}{3} \div 10$$

3. Perform Division: Evaluate $\frac{5}{3} \div 10$:

$$\frac{5}{3} \div 10 = \frac{5}{3} \times \frac{1}{10} = \frac{\overset{1}{\cancel{5}}}{3 \times \underset{2}{\cancel{10}}} = \frac{1}{6}$$

The expression becomes:

$$\frac{9}{4} + \frac{7}{6} - \frac{1}{6}$$

4. Perform Addition and Subtraction: The denominators are 4 and 6. The LCD is 12. Convert each fraction:

$$\frac{9}{4} = \frac{9 \times 3}{12} = \frac{27}{12}, \quad \frac{7}{6} = \frac{7 \times 2}{12} = \frac{14}{12}, \quad \frac{1}{6} = \frac{1 \times 2}{12} = \frac{2}{12}$$

Combine the terms:

$$\frac{27}{12} + \frac{14}{12} - \frac{2}{12} = \frac{27 + 14 - 2}{12} = \frac{39}{12}$$

Divide numerator and denominator by their greatest common divisor, 3:

$$\frac{39 \div 3}{12 \div 3} = \frac{13}{4} = 3\frac{1}{4}$$

Final Answer: $3\frac{1}{4}$

Question 4: Simplify

Expression: $1\frac{1}{2} \times 2\frac{3}{4} \div 1\frac{4}{7}$ **of** $2\frac{5}{8}$

1. Convert Mixed Numbers to Improper Fractions:

$$1\frac{1}{2} = \frac{3}{2}, \quad 2\frac{3}{4} = \frac{11}{4}, \quad 1\frac{4}{7} = \frac{11}{7}, \quad 2\frac{5}{8} = \frac{21}{8}$$

The expression becomes:

$$\frac{3}{2} \times \frac{11}{4} \div \frac{11}{7} \text{ of } \frac{21}{8}$$

2. Perform the “Of” Operation First: Evaluate $\frac{11}{7}$ of $\frac{21}{8}$:

$$\frac{11}{7} \text{ of } \frac{21}{8} = \frac{11}{7} \times \frac{21}{8} = \frac{11 \times \overset{3}{\cancel{21}}}{\underset{1}{\cancel{7}} \times 8} = \frac{33}{8}$$

The expression becomes:

$$\frac{3}{2} \times \frac{11}{4} \div \frac{33}{8}$$

3. Perform Division: The reciprocal of $\frac{33}{8}$ is $\frac{8}{33}$.

$$\frac{3}{2} \times \frac{11}{4} \div \frac{33}{8} = \frac{3}{2} \times \frac{11}{4} \times \frac{8}{33}$$

4. Multiply the Remaining Fractions: Using cancellation of common factors:

$$\frac{3 \times 11 \times 8}{2 \times 4 \times 33} = \frac{\overset{33}{\cancel{33}} \times 8}{8 \times \underset{33}{\cancel{33}}} = \frac{8}{8} = 1$$

Final Answer: 1

Question 5: Simplify

Expression: $\left(2\frac{3}{4} + 1\frac{5}{6}\right) \div 2\frac{1}{5}$ **of** $3\frac{1}{3}$

1. Convert Mixed Numbers to Improper Fractions:

$$2\frac{3}{4} = \frac{11}{4}, \quad 1\frac{5}{6} = \frac{11}{6}, \quad 2\frac{1}{5} = \frac{11}{5}, \quad 3\frac{1}{3} = \frac{10}{3}$$

The expression becomes:

$$\left(\frac{11}{4} + \frac{11}{6}\right) \div \frac{11}{5} \text{ of } \frac{10}{3}$$

2. Perform operations inside Bracket (B) First: Evaluate $\frac{11}{4} + \frac{11}{6}$. The LCD of 4 and 6 is 12:

$$\frac{11}{4} = \frac{11 \times 3}{12} = \frac{33}{12}, \quad \frac{11}{6} = \frac{11 \times 2}{12} = \frac{22}{12}$$

$$\frac{33}{12} + \frac{22}{12} = \frac{33 + 22}{12} = \frac{55}{12}$$

The expression becomes:

$$\frac{55}{12} \div \frac{11}{5} \text{ of } \frac{10}{3}$$

3. Perform the "Of" Operation: Evaluate $\frac{11}{5}$ of $\frac{10}{3}$:

$$\frac{11}{5} \text{ of } \frac{10}{3} = \frac{11}{5} \times \frac{10}{3} = \frac{11 \times \overset{2}{\cancel{10}}}{\cancel{5} \times 3} = \frac{22}{3}$$

The expression becomes:

$$\frac{55}{12} \div \frac{22}{3}$$

4. Perform Division: The reciprocal of $\frac{22}{3}$ is $\frac{3}{22}$.

$$\frac{55}{12} \div \frac{22}{3} = \frac{55}{12} \times \frac{3}{22} = \frac{\overset{5}{\cancel{55}} \times \overset{1}{\cancel{3}}}{\cancel{12}_4 \times \cancel{22}_2} = \frac{5 \times 1}{4 \times 2} = \frac{5}{8}$$

(Note: 55 and 22 were divided by their common factor 11; 3 and 12 were divided by 3.)

Final Answer: $\frac{5}{8}$

Question 6: Simplify

Expression: $\frac{7}{15}$ of $\left(\frac{2}{3} + \frac{7}{12}\right) \div \left(\frac{5}{6} - \frac{3}{5}\right)$

1. Evaluate the First Bracket $\left(\frac{2}{3} + \frac{7}{12}\right)$: The LCD of 3 and 12 is 12:

$$\frac{2}{3} = \frac{2 \times 4}{12} = \frac{8}{12} \implies \frac{8}{12} + \frac{7}{12} = \frac{8+7}{12} = \frac{15}{12}$$

Simplifying $\frac{15}{12}$ by dividing numerator and denominator by 3:

$$\frac{15}{12} = \frac{5}{4}$$

2. Evaluate the Second Bracket $\left(\frac{5}{6} - \frac{3}{5}\right)$: The LCD of 6 and 5 is 30:

$$\frac{5}{6} = \frac{5 \times 5}{30} = \frac{25}{30}, \quad \frac{3}{5} = \frac{3 \times 6}{30} = \frac{18}{30}$$

Subtracting the terms:

$$\frac{25}{30} - \frac{18}{30} = \frac{25-18}{30} = \frac{7}{30}$$

The expression becomes:

$$\frac{7}{15} \text{ of } \frac{5}{4} \div \frac{7}{30}$$

3. Perform the “Of” Operation: Evaluate $\frac{7}{15}$ of $\frac{5}{4}$:

$$\frac{7}{15} \text{ of } \frac{5}{4} = \frac{7}{15} \times \frac{5}{4} = \frac{7 \times \overset{1}{\cancel{5}}}{\underset{3}{\cancel{15}} \times 4} = \frac{7}{12}$$

The expression becomes:

$$\frac{7}{12} \div \frac{7}{30}$$

4. Perform Division: The reciprocal of $\frac{7}{30}$ is $\frac{30}{7}$.

$$\frac{7}{12} \div \frac{7}{30} = \frac{7}{12} \times \frac{30}{7} = \frac{\overset{1}{\cancel{7}} \times \overset{5}{\cancel{30}}}{\underset{2}{\cancel{12}} \times \underset{1}{\cancel{7}}} = \frac{5}{2} = 2\frac{1}{2}$$

(Note: 12 and 30 were divided by their common factor 6.)

Final Answer: $2\frac{1}{2}$

Question 7: Simplify

Expression: $\left(22 \div 5\frac{1}{2}\right) \div 2\frac{1}{5}$ of $3\frac{1}{3} + 1\frac{5}{11}$

Important Note on Printing Variations

In some textbooks, the operator inside the first bracket is printed as subtraction (−) instead of division (÷). We provide both solutions to match your printed version exactly.

Case A: If the expression is $(22 \div 5\frac{1}{2}) \div 2\frac{1}{5}$ of $3\frac{1}{3} + 1\frac{5}{11}$ (Standard BODMAS)

1. Convert Mixed Numbers to Improper Fractions:

$$5\frac{1}{2} = \frac{11}{2}, \quad 2\frac{1}{5} = \frac{11}{5}, \quad 3\frac{1}{3} = \frac{10}{3}, \quad 1\frac{5}{11} = \frac{16}{11}$$

The expression becomes:

$$\left(22 \div \frac{11}{2}\right) \div \frac{11}{5} \text{ of } \frac{10}{3} + \frac{16}{11}$$

2. Evaluate inside Bracket (B) First:

$$22 \div \frac{11}{2} = 22 \times \frac{2}{11} = \frac{\overset{2}{22} \times 2}{\underset{1}{11}} = 4$$

The expression simplifies to:

$$4 \div \frac{11}{5} \text{ of } \frac{10}{3} + \frac{16}{11}$$

3. Perform the “Of” Operation: Evaluate $\frac{11}{5}$ of $\frac{10}{3}$:

$$\frac{11}{5} \text{ of } \frac{10}{3} = \frac{11}{5} \times \frac{10}{3} = \frac{11 \times \overset{2}{10}}{\underset{1}{5} \times 3} = \frac{22}{3}$$

The expression becomes:

$$4 \div \frac{22}{3} + \frac{16}{11}$$

4. Perform Division: The reciprocal of $\frac{22}{3}$ is $\frac{3}{22}$.

$$4 \div \frac{22}{3} = 4 \times \frac{3}{22} = \frac{\overset{2}{4} \times 3}{\underset{11}{22}} = \frac{6}{11}$$

The expression becomes:

$$\frac{6}{11} + \frac{16}{11}$$

5. Perform Addition:

$$\frac{6}{11} + \frac{16}{11} = \frac{6+16}{11} = \frac{22}{11} = 2$$

Final Answer (Case A): 2

Case B: If the expression is $(22 - 5\frac{1}{2}) \div 2\frac{1}{5}$ of $3\frac{1}{3} + 1\frac{5}{11}$

1. Evaluate inside Bracket (Subtraction):

$$22 - 5\frac{1}{2} = \frac{22}{1} - \frac{11}{2} = \frac{44 - 11}{2} = \frac{33}{2}$$

The expression becomes:

$$\frac{33}{2} \div \frac{11}{5} \text{ of } \frac{10}{3} + \frac{16}{11}$$

2. Perform the "Of" Operation: As evaluated above, $\frac{11}{5}$ of $\frac{10}{3} = \frac{22}{3}$. The expression becomes:

$$\frac{33}{2} \div \frac{22}{3} + \frac{16}{11}$$

3. Perform Division:

$$\frac{33}{2} \div \frac{22}{3} = \frac{33}{2} \times \frac{3}{22} = \frac{\overset{3}{\cancel{33}} \times 3}{2 \times \underset{2}{\cancel{22}}} = \frac{9}{4}$$

The expression becomes:

$$\frac{9}{4} + \frac{16}{11}$$

4. Perform Addition (LCD of 4 and 11 is 44):

$$\frac{9 \times 11}{44} + \frac{16 \times 4}{44} = \frac{99}{44} + \frac{64}{44} = \frac{99 + 64}{44} = \frac{163}{44} = 3\frac{31}{44}$$

Final Answer (Case B): $3\frac{31}{44}$

Question 8: Simplify

Expression: $6\frac{1}{3} \div \left(2\frac{1}{5} + 3\frac{1}{2}\right)$ **of** $3\frac{1}{3}$

1. Convert Mixed Numbers to Improper Fractions:

$$6\frac{1}{3} = \frac{19}{3}, \quad 2\frac{1}{5} = \frac{11}{5}, \quad 3\frac{1}{2} = \frac{7}{2}, \quad 3\frac{1}{3} = \frac{10}{3}$$

The expression becomes:

$$\frac{19}{3} \div \left(\frac{11}{5} + \frac{7}{2}\right) \text{ of } \frac{10}{3}$$

2. Evaluate inside Bracket (Addition) First: Evaluate $\frac{11}{5} + \frac{7}{2}$. The LCD of 5 and 2 is 10:

$$\frac{11}{5} = \frac{11 \times 2}{10} = \frac{22}{10}, \quad \frac{7}{2} = \frac{7 \times 5}{10} = \frac{35}{10}$$

Adding the terms:

$$\frac{22}{10} + \frac{35}{10} = \frac{22 + 35}{10} = \frac{57}{10}$$

The expression simplifies to:

$$\frac{19}{3} \div \frac{57}{10} \text{ of } \frac{10}{3}$$

3. Perform the "Of" Operation: Evaluate $\frac{57}{10}$ of $\frac{10}{3}$:

$$\frac{57}{10} \text{ of } \frac{10}{3} = \frac{57}{10} \times \frac{10}{3} = \frac{\overset{19}{\cancel{57}} \times \overset{1}{\cancel{10}}}{\underset{1}{\cancel{10}} \times \underset{1}{\cancel{3}}} = 19$$

The expression becomes:

$$\frac{19}{3} \div 19$$

4. Perform Division: The reciprocal of 19 is $\frac{1}{19}$.

$$\frac{19}{3} \div 19 = \frac{19}{3} \times \frac{1}{19} = \frac{\overset{1}{\cancel{19}}}{3 \times \underset{1}{\cancel{19}}} = \frac{1}{3}$$

Final Answer: $\frac{1}{3}$